

Report R19 — the paper figures: the unitary census, $D_{\max}$, and how big the sectors actually are

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-08-13 (rev. 2; rev. 1 2026-08-12)

Abstract

This report accompanies `notebooks/paper_figures.ipynb`, an editable rebuild of R2’s Figure 1 — the number of Krylov sectors against N at `obc0` — together with the matching $D_{\max}$ figure and the distribution of sector sizes for the two exponentially fragmented rules W156 and W108. Revision 2 widens the scaling figures from five tracks to **all sixteen unitary rules**, ergodic ones included, which turns them from an illustration into a census: every rule built from I and V alone does exactly one of four things, and the two trivial behaviours — one sector of 2^N , or 2^N sectors of one — are the *frame* of both panels rather than omissions from them. They are also the only rules that saturate $n_{\text{wcc}} D_{\max} \geq 2^N$. The census is also *uniform*: the ergodic rules had been left at $N \leq 16$ by an early-exit in the sweep, and `extend_unitary` closed that gap with the flip reduction, so all sixteen rules now run to $N = 22$ on the same grid — 93 new units, each gated on the new code first reproducing every unit both sweeps had already stored for that rule. Four things are worth recording beyond the pictures. First, every curve is an **exact integer law**, checked at every computed N rather than fitted — 44 of the 48 (rule, series) slots, the other four being genuinely non-closed-form — so the annotations the figures carry are certificates (Table 2), and all of them survived the extension to $N = 22$. Second, **the vacuum alone decides whether some rules fragment**: reflection is an `obc0` symmetry and collapses sixteen rules to twelve curves, but the spin flip is not, and it puts the polynomially fragmented W60 and W102 opposite the completely ergodic W195 and W153. Third, W105 and W150 — routinely quoted as one entry “150/105” — differ at $N \equiv 1 \pmod{4}$ and at no other N , in *both* series at once, which is also the residue class at which W105 has no frozen state at all. Fourth, the size distribution separates the two exponential rules in a way neither scaling figure does: W156’s is **unimodal**, W108’s **monotone decreasing** with 33 552 singletons out of 170 625 sectors at $N = 21$, and in both the sector a random computational-basis state lands in is far smaller than $D_{\max}$.

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1 What this report is

R2 introduced the two scaling figures and R9 the growth-base plane built on them. What was missing was a version of the figures that can be *edited*: the package figure code in `scaling/figure.py` draws all 256 rules on a fixed template, which is the right tool for a sweep and the wrong one for a paper plate. `notebooks/paper_figures.ipynb` is the replacement. All the drawing code is inline in the notebook, all the adjustable parameters are collected in one cell at the top, and no cell calls the package's figure module.

What the notebook does *not* own is the arithmetic. The closed forms annotated on the curves live in `scaling/paper_figures.py`, and the notebook imports and re-verifies them before it draws anything. That split is deliberate: a law is a fact about the data, an axis limit is a preference, and only the second should be casually editable.

Outputs. `figures/paper/:` `fig_paper_sectors_obc0.pdf`, `fig_paper_dmax_obc0.pdf`, `fig_paper_scaling_obc0.pdf` (the two as panels (a) and (b) of one plate), `fig_paper_sizehist_obc0.pdf` and, new in this revision, `fig_paper_frozen_obc0.pdf`. Each with a 300 dpi PNG beside it.

2 The census: sixteen rules, four behaviours

A rule is unitary exactly when none of its four local channels is a reset, i.e. when its tuple lies in $\{I, V\}^4$; there are sixteen of them. Revision 1 of this report drew five of them. The other eleven were not missing because they are uninteresting but because of a loader: `summary.load_series` drops units flagged ergodic, which is correct when fitting a growth base and returns an *empty* series for seven rules. The notebook now uses its own loader and they are all present.

The four classes. Every unitary rule falls into exactly one, with no intermediate case anywhere in the computed range:

- **ergodic** — W51, W57, W99, W147, W153, W195 give literally one sector containing all 2^N states, at every computed N ; the six series are not merely similar but identical. W54 is the same with one exception: the all-zero state is frozen, because $r_{00} = I$, so it has 2 sectors of sizes $2^N - 1$ and 1.
- **polynomial** — W60, W102, W105, W150 fragment into $\Theta(N)$ sectors that are individually enormous. W60's largest is exactly 2^{N-1} , half the Hilbert space.
- **exponential** — W108, W156, W198, W201 fragment into a^N sectors with $1 < a < 2$.
- **frozen** — W204 is the identity circuit: 2^N sectors of one state each.

rule	tuple	class	refl.	flip	N	n_{wcc}	D_{max}	n_1	$n_{\text{wcc}}D_{\text{max}}/2^N$
51	VVVV	ergodic	51	51	6–22	1	4 194 304	0	1
54	IVVV	ergodic	54	147	6–22	2	4 194 303	1	2
57	VIVV	ergodic	99	99	6–22	1	4 194 304	0	1
99	VVIV	ergodic	57	57	6–22	1	4 194 304	0	1
147	VVVI	ergodic	147	54	6–22	1	4 194 304	0	1
153	VIVI	ergodic	195	102	6–22	1	4 194 304	0	1
195	VVII	ergodic	153	60	6–22	1	4 194 304	0	1
60	IIVV	polynomial	102	195	6–22	23	2 097 152	2	11.5
102	IVIV	polynomial	60	153	6–22	23	2 097 152	2	11.5
105	VIIV	polynomial	105	105	6–22	12	1 352 078	1	3.868
150	IVVI	polynomial	150	150	6–22	12	1 352 078	1	3.868
108	IIIV	exponential	108	201	6–22	299 426	17 711	54 289	1264
156	IIVI	exponential	198	198	6–22	46 368	576	12	6.368
198	IVII	exponential	156	156	6–22	46 368	576	12	6.368
201	VIII	exponential	201	108	6–22	170 625	46 368	20 736	1886
204	IIII	frozen	204	204	6–22	4 194 304	1	4 194 304	1

Table 1: The unitary census at **obc0**: all sixteen rules, the reflection and spin-flip partner of each, and the three counts at the largest computed N . The last column is the hyperbola $n_{\text{wcc}}D_{\text{max}} \geq 2^N$; it equals one exactly for the six ergodic rules and for the identity, the only rules whose sectors all have the same size, and exceeds it by up to three orders of magnitude for the fragmented ones. Reflection preserves the series; the spin flip does not, and it is what puts an ergodic rule opposite a fragmented one in the pairs (60, 195) and (102, 153).

The trivial classes are the frame, not the margin. The hyperbola $n_{\text{wcc}}D_{\text{max}} \geq 2^N$ (R16) holds with equality exactly when every sector has the same size. Two behaviours manage that — one sector of 2^N and 2^N sectors of one — and they are precisely the two that do not fragment. The last column of Table 1 is that ratio: exactly 1 for the six ergodic rules and for the identity, 2 for W54, and up to 1.3×10^3 for W201. So the ergodic and identity curves in Figure 1 are not extra data points on the same footing as the rest; they are the boundary of the region, and fragmentation is what happens strictly inside it.

3 Two symmetries, one of which the boundary breaks

Sixteen rules do not give sixteen curves, and the reason is worth stating precisely because the two candidate symmetries behave oppositely at **obc0**.

No symmetry is a source of data. Reflection *is* a symmetry of these series, and the sweep driver offers a representatives-only mode, so “W198 = W156” could in principle have been a substitution rather than a measurement. It is not, and the report does not ask the reader to take that on trust. Every unitary rule carries its own records with its own runtimes — W156 and W198 took 217 and 199 seconds at $N = 21$, W60 and W102 took 6982 and 6608 — and `tests/test_extend_unitary.py` recomputes *every* stored unit of *every* one of the sixteen rules from that rule’s own gate assignment, by label propagation on its own flip graph, consulting no partner, and requires it to reproduce what both stores hold. Three unrelated algorithms, rule by rule, 33 units each. The identities below are therefore results, and the figures draw both members of a pair from their own data rather than mirroring one onto the other.

Reflection ($r_{01} \leftrightarrow r_{10}$) is a symmetry. It leaves the sector decomposition invariant for unitary rules at both boundary conventions, so the sixteen rules give **twelve** distinct series and

the pairs $\{57, 99\}$, $\{60, 102\}$, $\{153, 195\}$, $\{156, 198\}$ agree term for term. Table 2 is organised by these twelve classes.

The spin flip ($0 \leftrightarrow 1$ on every site) is not. $X^{\otimes N}$ is a basis permutation and XHX has the same matrix magnitudes, so it preserves the transition graph — under pbc. At obc0 the padding is pinned to $|0\rangle$ and the flip sends it to $|1\rangle$. The consequence is sharp: of the six spin-flip pairs, the only two whose series match are $\{57, 99\}$ and $\{156, 198\}$, which are *also* reflection pairs. The flip never preserves a series on its own account (`flip_survives_only_via_reflection`), and in two of the four remaining pairs it does something much stronger than shift a curve:

$$\text{W60 (IIVV)} : N + 1 \text{ sectors} \quad \longleftrightarrow \quad \text{W195 (VVII)} : 1 \text{ sector},$$

and likewise $\text{W102} \leftrightarrow \text{W153}$. Same gate set, same lattice, same brickwork; only the vacuum differs, and one of the pair fragments while the other thermalises completely. The remaining two pairs stay within their class: W54 against W147 differ by exactly the one frozen vacuum state, and W108 against W201 is the familiar one-site shift.

4 Every curve is an exact law

rules	tuple	class	n_{wcc}	D_{max}	frozen n_1	N	holds
51	VVVV	ergodic	1	2^N	0	6–22	yes
57, 99	VIVV	ergodic	1	2^N	0	6–22	yes
147	VVVI	ergodic	1	2^N	0	6–22	yes
153, 195	VIVI	ergodic	1	2^N	0	6–22	yes
54	IVVV	ergodic	2	$2^N - 1$	1	6–22	yes
60, 102	IIVV	polynomial	$N + 1$	2^{N-1}	2	6–22	yes
105	VIIIV	polynomial	$\lfloor N/2 \rfloor + 1 + [N \equiv 3 \pmod{4}]$	$\max_w \binom{N+1}{w}$, odd w at $N \equiv 1 \pmod{4}$	1, 2, 1, 0 for $N \equiv 0, 1, 2, 3 \pmod{4}$	6–22	yes
150	IVVI	polynomial	$\lfloor (N+1)/2 \rfloor + 1$	$\max_{w \text{ even}} \binom{N+1}{w}$	1 (N even), 2 (odd)	6–22	yes
108	IIIV	exponential	—	F_N	$F_{\lfloor N/2 \rfloor + 2} F_{\lceil N/2 \rceil + 2}$	6–22	yes
156, 198	IIVI	exponential	F_{N+2}	—	$\lfloor (N+1)/2 \rfloor + 1$	6–22	yes
201	VIII	exponential	—	F_{N+2}	F_{m+1}^2 ($N=2m$), $F_m F_{m+3}$ ($N=2m+1$)	6–22	yes
204	IIII	frozen	2^N	1	2^N	6–22	yes

Table 2: Every unitary rule at obc0, one row per reflection class (reflection is an obc0 symmetry, so the sixteen rules give twelve distinct series). Each closed form was read off the integer series and then evaluated at *every* computed N ; the last column is that check, not a goodness of fit. “—” marks a series with no exact law: W156/W198 ’s largest sector mixes rooms of two lengths and only approaches $4^{N/5}$, and W108/W201 ’s sector counts satisfy a recurrence rather than a closed form. F_k is the Fibonacci number with $F_1 = F_2 = 1$ and $m = \lfloor N/2 \rfloor$.

None of the entries in Table 2 is a regression. Each was read off the integer series and then evaluated at every N in the sweep — all seventeen points, $N = 6 \dots 22$, for every rule — and the last column reports that comparison, not a residual. `scaling/paper.figures.verify` performs it, the notebook asserts on it before drawing, and `tests/test_paper_figures.py` runs it independently of either. Of the $16 \times 3 = 48$ (rule, series) slots, 44 carry an exact closed form; the four that do not are W156/W198 ’s D_{max} and W108/W201 ’s sector count, and the test suite asserts that the set of gaps is exactly those four, so a curve cannot quietly lose its certificate.

Those four are covered by recurrences and an asymptote, checked the same way:

- W108’s sector count satisfies $a_N = 2a_{N-1} - a_{N-2} + a_{N-3}$ exactly, whose dominant root is $\rho^2 = 1.754878\dots$, the square of the plastic number. R9 obtained that base by fitting and R18 derived it from the wall grammar; the recurrence is a third and purely arithmetic route to it.
- W201’s sector count is W108’s evaluated one site shorter, $n_{\text{wcc}}^{201}(N) = n_{\text{wcc}}^{108}(N - 1)$, at every overlapping N .
- W156’s D_{max} tag reads $4^{N/5}$ and is deliberately *not* a law. At finite N the largest sector is a mixture of rooms of length 2 and 3 (R2 sec. 3), so no exact recurrence holds over the computed range. What is reported instead is the five-step ratio, which needs no fit: $(D_{\text{max}}(21)/D_{\text{max}}(16))^{1/5} = 1.3195079$ against $4^{1/5} = 1.3195079$. Six digits by $N = 21$ is evidence for the derived base; an exact-law assertion would be the wrong claim to make.

4.1 150/105 is not one entry, and 60/102 is

The two polynomial tracks are usually written as pairs, but they are not pairs in the same sense.

W60 and W102 agree *term for term* in both series, so the figure labels that track with an equals sign.

W105 and W150 do not. At $N \equiv 1 \pmod{4}$ — and, over the whole computed range, at no other N — W105’s largest sector carries **odd** domain-wall number where W150’s carries even ($N = 9$: $\binom{10}{5} = 252$ against $\binom{10}{4} = 210$), and at exactly those N W105 has one sector fewer. It is one phenomenon appearing in both panels: an extra accessible level of the wall number is precisely one fewer sector. The figure therefore labels that track with a comma and draws it as two curves — the second rule as a larger open marker underneath the first, so that where the two coincide the reader sees a filled mark inside a ring and where they part sees a lone ring. The four lone rings in the sector panel are at $N = 9, 13, 17, 21$. Section 8 adds a third face of the same parity obstruction.

5 Figures 1 and 2 — the scaling

The two panels are also available separately as `fig_paper_sectors_obc0.pdf` (Figure 2) and `fig_paper_dmax_obc0.pdf` (Figure 3), at single-column proportions, for a paper that wants them apart.

Where each point comes from. Not every rule had been pushed equally far. The Tier-1a sweep reaches $N = 21$ on the fragmented rules and 22 for W150, but stops enlarging N as soon as a rule is classified ergodic, so those seven ended at 13 there; the Tier-1e sweep covered all 256 rules to $N = 16$ (a few to 17). Since weak and strong components coincide for unitary rules the two stores are merged, and `wcc_scc_agreement` checks every overlapping unit before the merge is used — there is no disagreement anywhere in the sixteen.

That still left the census ragged, for a reason having nothing to do with the rules, so `qca_fragmentation/extend_unitary.py` closed it (Section 6). Each curve is nonetheless drawn only over its own range and never extrapolated, and the machinery that annotates a short track’s N_{max} in the legend is still in place, because the ranges can diverge again the moment anyone pushes one rule further.

On the drawing. The y axis is a true logarithmic axis with decade ticks rather than $\ln(\cdot)$ on a linear axis, as in the original R2 figures. A straight line still means an exponential, and the reader additionally gets to read the actual sector counts off the axis. The five hues are the CVD-validated categorical order used across the reports, re-checked over all pairs under normal, protan, deutan and tritan vision; the worst separation is $\Delta E = 7.4$ (W201 green against W60

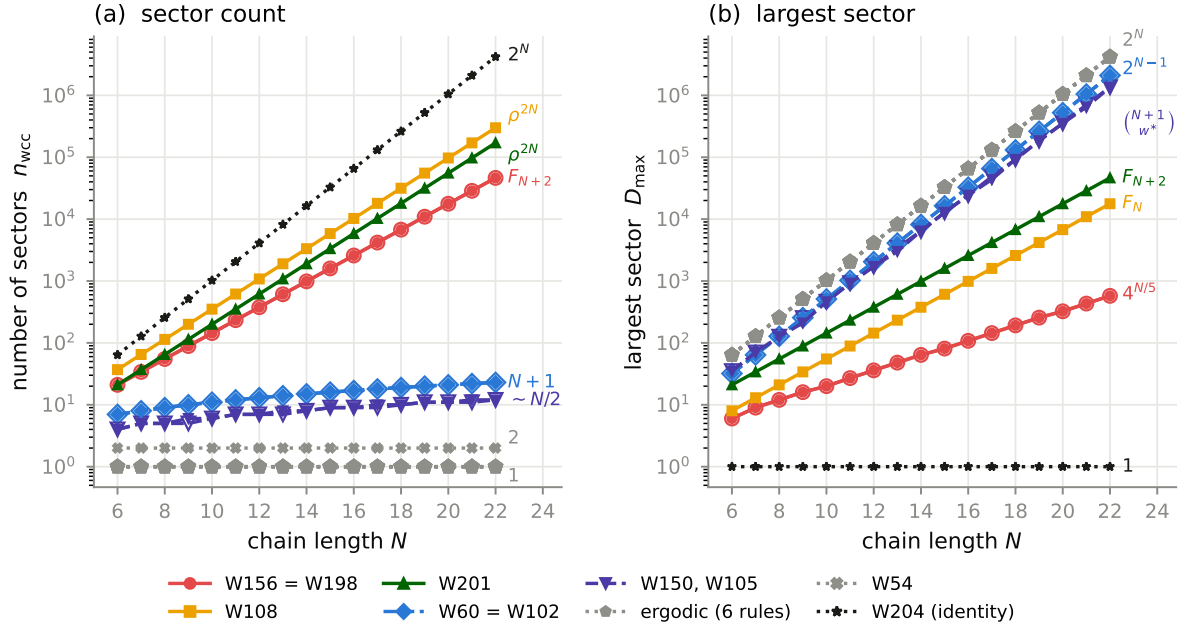


Figure 1: Sector count (a) and largest sector (b) against chain length for all sixteen unitary rules (`obc0`, $\log y$). Four rules fragment exponentially — W156=W198 into F_{N+2} sectors, W108 and W201 into $\Theta(\rho^{2N})$ — while W60/W102 and W150/W105 produce only $\Theta(N)$ sectors and correspondingly huge ones: W60’s largest is exactly 2^{N-1} , half the entire Hilbert space. The grey and black tracks are the trivial extremes that bound both panels: six ergodic rules with one sector of 2^N (W54: plus one frozen vacuum state), and the identity W204 with 2^N sectors of one. Every tag is an exact law (Table 2) except W156’s $4^{N/5}$, the derived asymptote. Panels (a) and (b) are near mirror images because $n_{\text{wcc}} \cdot D_{\text{max}} \geq 2^N$, and the trivial rules are exactly the ones for which that holds with equality. All sixteen rules span the same grid $N = 6 \dots 22$.

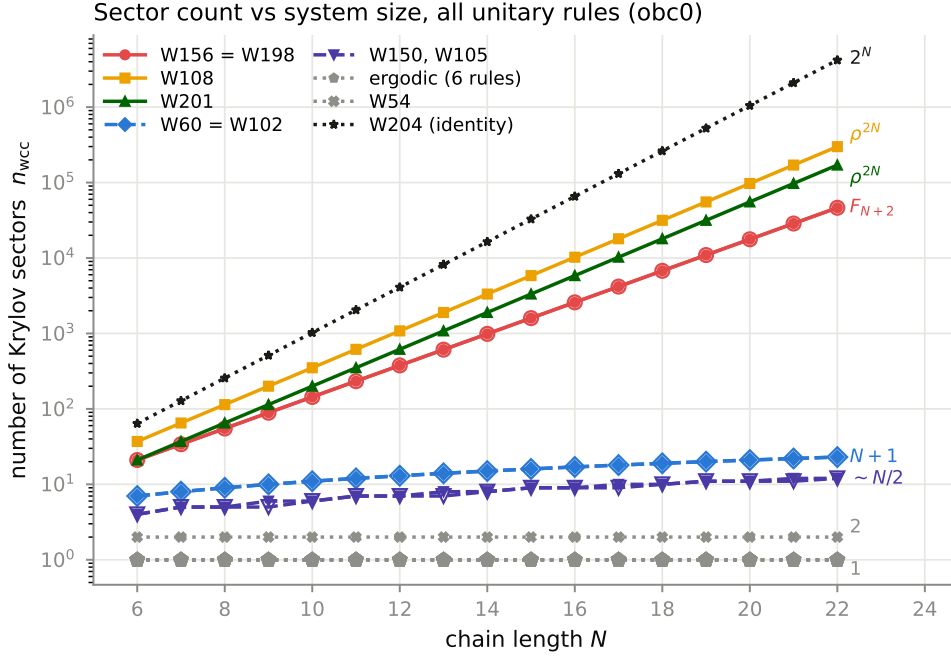


Figure 2: Number of Krylov sectors vs N (obc0), log scale — the R2 Figure 1 content, rebuilt and completed. Exponential fragmentation (156=198, 108, 201) against the linear and central-binomial families (60/102, 150/105), with the ergodic rules on the floor $n_{\text{wcc}} = 1$ and the identity on the ceiling $n_{\text{wcc}} = 2^N$.

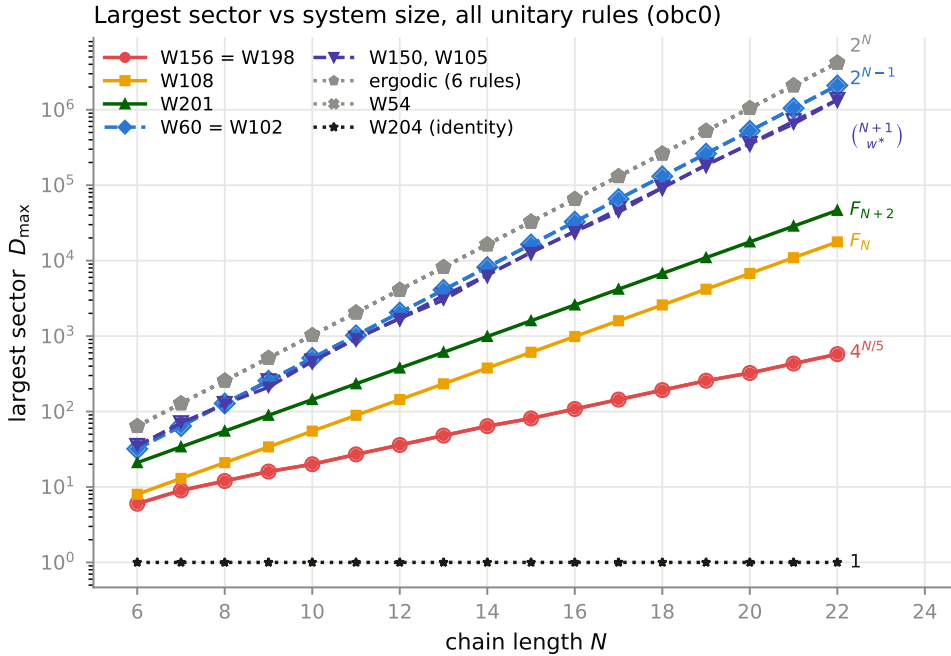


Figure 3: Largest sector D_{max} vs N (obc0), log scale. The grey track is the 2^N ceiling, traced exactly by the six ergodic rules (W54 one state below it). The ordering is the reverse of Figure 2: the rules with the fewest sectors have the biggest ones.

blue, tritanopia), which is admissible only with a secondary encoding, and three are present — distinct markers, a line style that tracks the class, and a direct law tag at every line end. The green was darkened from #008300 to #006400 in this revision, which raises that worst pair from 3.9 and improves every other pair it takes part in. The two trivial classes are drawn in neutrals rather than a sixth and seventh hue, which is a statement about the content: they are the reference lines of the figure, and the saturated hues belong to the rules that actually fragment.

6 Closing the N gap: `extend_unitary`

The seven rules the sweeps stopped early on are the cheapest of the sixteen, not the dearest. For a *unitary* rule the sector partition is exactly the connected-component structure of the single-flip graph $x \sim x \oplus 2^i$ over the sites where the Hadamard fires — a theorem, and one validated against the streamed engine for all sixteen unitary rules at both boundary conventions in `tests/test_flip_graph.py`. Label propagation over one `uint32` array of 2^N entries decides it: $N = 22$ is 17 MB and a quarter of a second per rule.

`extend_unitary` does that for the missing units and appends them to the Tier-1e store in the ordinary schema, with `checks.source` recording where they came from. It writes nothing for a rule until the same code has reproduced *every* unit both stores already hold for it — between 19 and 28 units per rule, computed by union-find in one store and by Tarjan in the other, so three unrelated algorithms have to agree before a fourth number is added. 93 units were added this way, taking every unitary rule to $N = 22$. The whole run takes well under a minute, most of it spent re-deriving the units that already existed rather than on the new ones; run again it adds nothing and re-verifies all 33 units of each of the sixteen rules in half a minute.

Two things fell out of it. Every law in Table 2 continues to hold at the new N , which is a real test rather than a restatement — the extension could have broken any of the 44. And the fast path reproduces the Tier-1a record for W150 at $N = 22$, which is the deepest unit in the whole Tier-1a archive at either boundary condition — every other rule stops at 21 — so the check reaches the archive’s own frontier.

The same command extends any unitary rule further: `python -m qca_fragmentation.extend_unitary --n-max 24`. Memory is $4 \cdot 2^N$ bytes, so $N = 26$ is 268 MB and $N = 30$ is 4.3 GB; the reduction is proved only for unitary rules and refuses anything else.

7 Figure 3 — how big are the sectors?

Neither scaling figure says anything about the *distribution* of sector sizes: n_{wcc} counts them and D_{max} reports one extreme. Nor does either separate the two exponentially fragmented rules from each other — both are straight lines of similar slope. Figure 4 does both, and the two columns turn out to look nothing alike.

Two different kinds of exponential fragmentation. W156 and W108 sit close together in the growth-base plane — both exponential in both coordinates — and Figure 4 shows that this hides a qualitative difference. W156’s sectors are all of comparable size: at $N = 21$ its 28 657 sectors have mean 73 and median 108, and only 12 of them are singletons. W108’s are not: the same N gives 170 625 sectors of which 33 552, nearly a fifth, are frozen single states, the distribution falls monotonically, and the median is 42 against a D_{max} of 10 946.

D_{max} overstates the typical sector, increasingly. The ratio $D_{\text{max}}/\text{median}$ runs $2.7 \rightarrow 3.2 \rightarrow 4.0$ for W156 over $N = 13, 17, 21$ and $29 \rightarrow 76 \rightarrow 261$ for W108. Both grow, so the gap is not a small- N artefact. This matters wherever D_{max} stands in for “how much of the Hilbert

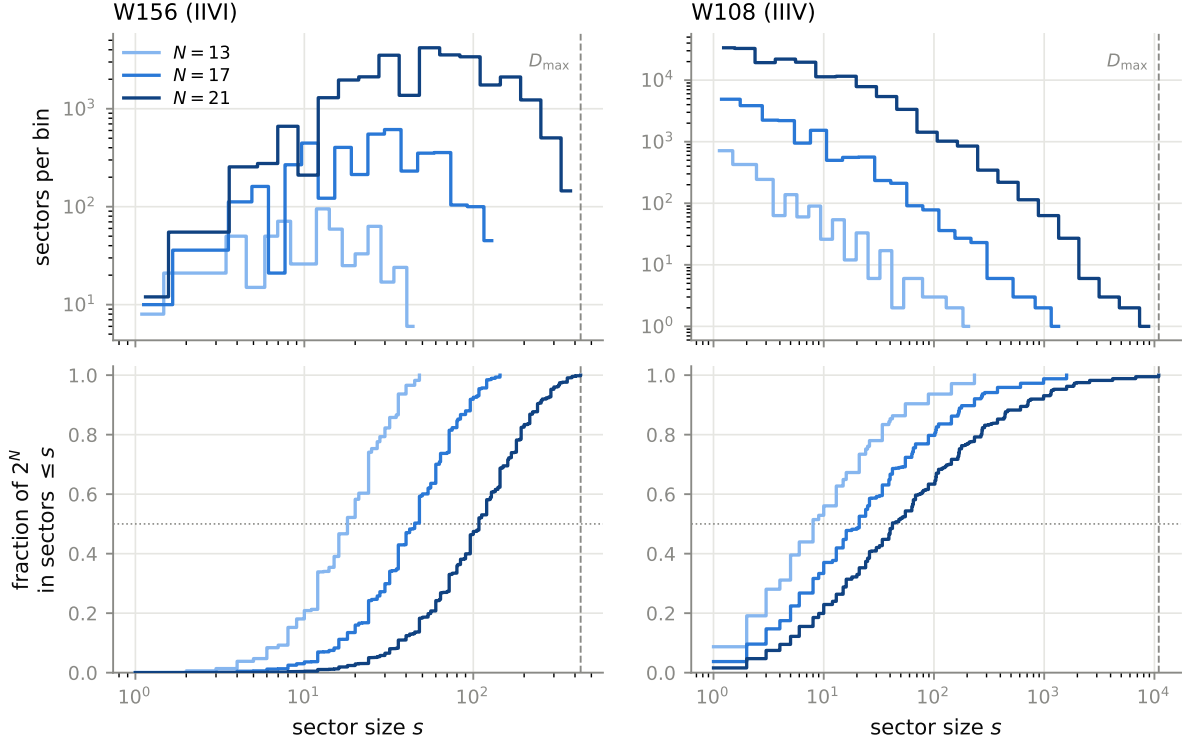


Figure 4: Sector-size distribution of W156 (left) and W108 (right) at $N = 13, 17, 21$ (obc0), light to dark. **Top:** number of sectors per log-spaced size bin. W156 is unimodal and peaked well away from $s = 1$; W108 is monotone decreasing over four decades. **Bottom:** the fraction of the 2^N computational-basis states lying in sectors of size $\leq s$ — read the median sector size off where a curve crosses the dotted line at $1/2$. The dashed vertical rule is $D_{\max}$ at $N = 21$; in both rules the typical sector is far to the left of it. The sawtooth in the top-left panel is real and not noise: at small s a bin holds one or two integer sizes, and even sizes are much commoner than odd.

rule	N	sectors	$D_{\max}$	$D_{\max}/2^N$	mean	median	singletons
156	13	610	48	5.86e-03	13.4	18	8
156	17	4 181	144	1.10e-03	31.3	45	10
156	21	28 657	432	2.06e-04	73.2	108	12
108	13	1 897	233	2.84e-02	4.3	8	714
108	17	17 991	1 597	1.22e-02	7.3	21	4 895
108	21	170 625	10 946	5.22e-03	12.3	42	33 552

Table 3: The numbers behind Figure 4 (obc0). *mean* is $2^N/n_{\text{wcc}}$; *median* is the size of the sector a uniformly random computational-basis state lands in, i.e. where the cumulative curve crosses $1/2$. Both are far below $D_{\max}$, which is why the largest sector is a poor description of the typical one.

space is dynamically connected”: for W108 at $N = 21$ the largest sector holds 0.5% of the space, but a state drawn uniformly at random sits in a sector of about 42 states out of two million.

Why this can be computed at all. The stored `sizes_recurrent` list is truncated at 2048 entries for these rules once $N \geq 16$ (W108: $N \geq 14$), which would cap Figure 4 at $N = 15$, or 13 for W108. The `size_hist` field is *not* truncated — it carries every class — and `size_stats` asserts both that its class count equals n_{wcc} and that $\sum_s s h(s) = 2^N$ before returning, so relying on it is checked rather than assumed. That is what puts $N = 21$ within reach.

8 Figure 4 — the frozen states

Certifying the census produced a third exact series for free. The number of size-one sectors is a closed form for *every* unitary rule, and it is the sharpest single number for telling the four classes apart.

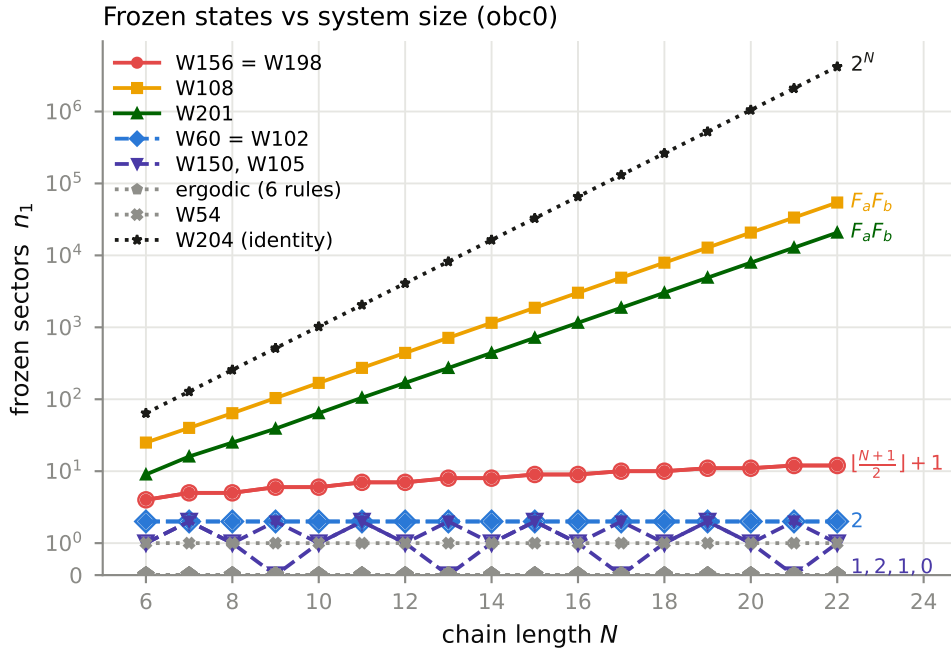


Figure 5: Number of frozen states n_1 vs N (obc0), symlog y so that zero is a value rather than a gap — which matters twice here: the six ergodic rules have none at all, and W105 has none at $N \equiv 1 \pmod{4}$. Exponential rules have exponentially many (W108: a third of a million by $N = 21$), the polynomial rules have $O(1)$, and the identity has all 2^N .

Two of the laws are worth pulling out of Table 2.

W108’s frozen count factorises over the sublattices.

$$n_1^{108}(N) = F_{\lfloor N/2 \rfloor + 2} F_{\lceil N/2 \rceil + 2},$$

exactly, at every computed N — 25, 40, 64, 104, 169, ... up to $54\,289 = F_{13}^2$ at $N = 22$. A state is frozen iff nothing can move on either sublattice, and the two decouple, so the count is a product of two Fibonacci numbers of nearly equal index. That is the same even/odd split that carries R18’s two independent domain-wall charges $|W_{\text{even}}|$ and $|W_{\text{odd}}|$, seen here in a completely different observable. W201 obeys the same form with the indices pulled down by one: F_{m+1}^2 at $N = 2m$ and $F_m F_{m+3}$ at $N = 2m + 1$, the odd case being W108’s value at $N - 2$ off by $(-1)^m$ via Catalan’s identity.

W105 loses its frozen states exactly where it parts from W150. n_1^{105} runs 1, 2, 1, 0 as N runs through 2, 3, 0, 1 (mod 4), so the zeros fall at $N \equiv 1 \pmod{4}$ — precisely the residue class at which W105 has one sector fewer than W150 and its largest sector jumps to odd domain-wall number. W150 never drops below one. Three observables, one parity obstruction.

9 Using and editing the notebook

Open `notebooks/paper_figures.ipynb` and run all cells; it writes into `figures/paper/` and takes a few seconds. Section 2 of the notebook holds every adjustable parameter — page widths, font sizes, the palette, the **TRACKS** list that decides which rules appear and how they are grouped, the law tags and their nudges, the y scale per series, the legend placement, the histogram’s N values and bin count, the output directory. The drawing code reads that cell and nothing else.

To add or regroup rules in the scaling figures, edit **TRACKS**; rules inside one track share a hue and the leading one is drawn filled on top of the others, so a track that is secretly two curves gives itself away. To change which N the histograms show, edit **HIST_NS**. To change a *law*, edit `scaling/paper_figures.py` instead — and expect the test suite to have an opinion. The notebook itself is generated by `notebooks/_build_paper_figures_notebook.py`, so a hand edit that should survive belongs in the builder.

10 Caveats

- **obc0 only.** The **pb**c sweep is held pending caveats (`NEXT_STEPS.md`); under **pb**c at odd N the brick-wall layers stop commuting, so those points would need even N only. This matters more than usual for Section 3: the ergodic-versus-fragmented contrasts it reports are *created* by the **obc0** vacuum, and under **pb**c the spin flip is restored and the pairs would collapse onto each other.
- **The tail of every rule comes from the flip reduction**, not from the sweep that produced its head: $N \geq 17$ (or 18) was computed by `extend_unitary`, and those records say so in `checks.source`. The reduction is a theorem for unitary rules and for no others, it is checked against the engine independently, and the extension reproduces every stored unit of a rule before adding to it — but it is a different algorithm from the one that produced the older points, and that is worth knowing.
- **These are WCC sectors**, which for the unitary rules shown coincide with the strongly connected components, so the decomposition is exact for the monitored and the unmonitored chain alike. That coincidence is a property of unitarity and does not survive dissipation — see R-T14 on the recurrent-class/WCC distinction. It is also what licenses merging the Tier-1a and Tier-1e sweeps.
- **Both members of a reflection pair are measured and drawn**, as a filled marker inside a larger open one, rather than one being computed and the other inferred; a pair that ever stopped coinciding would show as a lone ring. Nothing plotted anywhere in this report is a symmetry image of another number.
- **Nothing here is new dynamics.** The sweep data is R2’s and Tier-1e’s, extended along the same N axis; what is new is the completion of the census, the certification of the laws including the frozen-state series, the spin-flip statement, and the size distributions.